1. Suppose 111 points are given within an equilateral triangle of side 15. Prove that it is always possible to cover at least 3 of these points by a round coin of diameter $\sqrt{3}$ (part of which may lie outside the triangle).

2. Let $a_1, a_2, \ldots, a_n$ be arbitrary real numbers. Prove that there exists a real number x such that each of the numbers $x + a_1, x + a_2, \ldots, x + a_n$ is irrational.

3. A positive integer k greater than 1 is given. Prove that there exist a prime p and a strictly increasing sequence of positive integers $a_1, a_2, \ldots$ such that the terms of the sequence $p + ka_1, p + ka_2, \ldots$ are all primes.

4. Suppose $\{a_n\}_{n=1}^{\infty}$ is a sequence of integers which satisfies the recursion $a_n = Pa_{n-1} + Qa_{n-2}$ where $P, Q \in \mathbb{Z}$. Prove that for any integer k the sequence of remainders obtained when a_n's are divided by k is periodic.

5. Prove that for any integer $k > 0$ the Fibonacci sequence contains infinitely many multiples of k.

6 (1985 IMO)**.** M be set of positive integers such that prime divisors of elements of M are from the set $\{p_1, \ldots, p_n\}$. Prove that if $|M| > 3.2^n$ then there exists $a, b, c, d \in M$ pairwise distinct such that $abcd$ is a perfect fourth power.

7. Let $P_1, P_2, \ldots, P_{2n}$ be a permutation of the vertices of a regular polygon. Prove that $P_1P_2, P_2P_3, \ldots, P_{2n-1}P_{2n}, P_{2n}P_1$ contains a pair of parallel lines.

8 (1999 Poland)**.** Let $0 \leq a_1 < a_2 < \cdots < a_{101} < 5050$ be integers. prove that there exists a_i, a_j, a_k, a_l distinct such that $5050 | a_i + a_j - a_k - a_l$.

9 (Great Britain 1976)**.** Let $A_1, A_2, \ldots, A_{50}$ subsets of a finite set A such that any subset has more than half of the number of elements of A. Prove that there exists a subset of A with at most 5 elements that has non-empty intersection with each of the 50 subsets.

10. Let $A = \{1, \ldots, 100\}$ and let $A_1, A_2, \ldots, A_m$, be subsets of A, each with 4 elements, any two of them having at most 2 elements in common. Prove that if $m \geq 40425$ then there exist 49 subsets among the chosen ones such that their union is A, but the union of any 48 subsets (among the 49) is not A.

11 (Mantel's Theorem)**.** Consider a set of $2n$ points in space, $n > 1$. No three are collinear. Prove that maximum n^2 segments (with end points from the given points) can be drawn without forming any triangle with vertices from the given points.

12. Six points are given in space such that the pairwise distances between them are all distinct. Consider the triangles with vertices at these points. Prove that the longest side in one of these triangles is at the same time the shortest side in another triangle.

13 (Putnam 2000)**.** Let B be a set of more than $\frac{2^{n+1}}{n}$ distinct points in n-dimensional space with coordinates of the form $(\pm 1, \ldots, \pm 1)$, where $n \geq 3$. Show that there are 3 distinct points in B which are the vertices of an equilateral triangle.

14 (1987 Chinese TST)**.** There are $2n$ distinct points in space, where $n \geq 2$. No four of them are on the same plane. If $n^2 + 1$ pairs of them are connected by line segments, then prove that there are two triangles sharing a common side.

15 (1989 Chinese Team Training)**.** There are $2n$ distinct points in space, where $n \geq 2$. No four of them are on the same plane. If $n^2 + 1$ pairs of them are connected by line segments, then prove that there are at least n distinct triangles formed.

16. Suppose two players, first A and then B, take turns writing down positive integers subject to the two rules

(i) no integer may exceed an agreed upon limit L, and

(ii) no integer may be a divisor of a number already used.

The first one unable to play is the loser. Prove that A has a winning strategy.

17 (1985 Bulgarian Olympiad)**.** Let $P_1, P_2, \ldots, P_7$ be any 7 points in space, no four in the same plane, and suppose that for each of the $\binom{7}{2} = 21$ segments P_iP_j is coloured with either red or blue. Prove that, no matter how the colours are distributed, two monochromatic triangles will always be formed which have no side in common.

18. Given $2n$ points in a plane with no three of them collinear. Show that they can be divided into n pairs such that the n segments joining each pair do not intersect.

19 (1974 Kiev Mathematical Olympiad)**.** Numbers $1, 2, 3, \ldots, 1974$ are written on a board. You are allowed to replace any two of these numbers by one number, which is either the sum or the difference of these numbers. Show that after 1973 times performing this operation, the only number left on the board cannot be 0.

20. In an 8×8 board, there are 32 white pieces and 32 black pieces, one piece in each square. If a player can change all the white pieces to black and all the black pieces to white in any row or column in a single move, then is it possible that after finitely many moves, there will be exactly one black piece left on the board?

21. Four x's and five o's are written around the circle in an arbitrary order. If two consecutive symbols are the same, then insert a new x between them. Otherwise insert a new o between them. Remove the old xs and os. Keep on repeating this operation. Is it possible to get nine os?

22. There are three piles of stones numbering 19, 8 and 9, respectively. You are allowed to choose two piles and transfer one stone from each of these two piles to the third piles. After several of these operations, is it possible that each of the three piles has 12 stones?

23. Two boys play the following game with two piles of candies. In the first pile, there are 12 candies and in the second pile, there are 13 candies. Each boy takes turn to make a move consisting of eating two candies from one of the piles or transferring a candy from the first pile to the second. The boy who cannot make a move loses. Show that the boy who played second cannot lose. Can he win?

24. Each member of a club has at most three enemies in the club. (Here enemies are mutual.) Show that the members can be divided into two groups so that each member in each group has at most one enemy in the group.

25 (1961 All-Russian Math Olympiad)**.** Real numbers are written in an $m \times n$ table. It is permissible to reverse the signs of all the numbers in any row or column. Prove that after a number of these operations, we can make the sum of the numbers along each line (row or column) non-negative.

26. There is a table with a square top. Two players take turn putting a dollar coin on the table. The player who cannot do so loses the game. Show that the first player can always win.

27 (Bachets Game)**.** Initially, there are n checkers on the table, where $n > 0$. Two persons take turn to remove at least 1 and at most k checkers each time from the table. The last person who can remove any checker wins the game. For what values of n will the first person have a winning strategy? For what values of n will the second person have a winning strategy?

28 (Game of Nim)**.** There are 3 piles of checkers on the table. The first, second and third piles have x, y and z checkers respectively in the beginning, where $x, y, z > 0$. Two persons take turn choosing one of the three piles and removing at least one to all checkers in that pile each time from the table. The last person who can remove any checker wins the game. Who has a winning strategy?

29. Twenty girls are sitting around a table and are playing a game with n cards. Initially, one girl holds all the cards. In each turn, if at least one girl holds at least two cards, one of these girls must pass a card to each of her two neighbours. The game ends if and only if each girl is holding at most one card.

(a) Prove that if $n \geq 20$, then the game cannot end.

(b) Prove that if $n < 20$, the game must end eventually.

30 (1996 Irish Math Olympiad)**.** On a 5×9 rectangular chessboard, the following game is played. Initially, a number of discs are randomly placed on some of the squares, no square containing more than one disc. A turn consists of moving all of the discs subject to the following rules:

(i) each disc may be moved one square up, down, left or right;

(ii) if a disc moves up or down on one turn, it must move left or right on the next turn, and vice versa;

(iii) at the end of each turn, no square can contain two or more discs.

The game stops if it becomes impossible to complete another turn. Prove that if initially 33 discs are placed on the board, the game must eventually stop. Prove also that it is possible to place 32 discs on the board so that the game can continue forever.

31 (1995 Israeli Math Olympiad)**.** Two players play a game on an infinite board that consists of 1×1 squares. Player I chooses a square and marks it with an O. Then, player II chooses another square and marks with an X. They play until one of the players marks a row or a column of five consecutive squares, and this player wins the game. If no player can achieve this, the game is a tie. Show that player II can prevent player I from winning.

32 (1999 USAMO)**.** The Y2K Game is played on a 1×2000 grid as follow. Two players in turn write either an S or an O in an empty square. The first player who produces three consecutive boxes that spell SOS wins. If all boxes are filled without producing any SOS, then the game is a draw. Prove that the second player has a winning strategy.

33 (1993 IMO)**.** On an infinite chessboard, a game is played as follow. At the start, n^2 pieces are arranged on the chessboard in an $n \times n$ block of adjoining squares, one piece in each square. A move in the game is a jump in a horizontal or vertical direction over an adjacent occupied square to an unoccupied square immediately beyond. The piece that has been jumped over is then removed. Find those values of n for which the game can end with only one piece remaining on the board.

34. At a party, no boy dances with every girl, but each girl dances with atleast one boy. Prove that there exists two pairs (b_1, g_1) and (b_2, g_2) of boy and girl who dance, where as b_1 does not dance with g_2 nor b_2 does dance with g_1.

35. In the sequence $1, 0, 1, 0, 1, 0, \ldots$ each term, starting with the seventh is equal to the last digit of the sum of preceding 6 numbers. Prove that this sequence doesn't contain 6 consecutive terms equal to $0, 1, 0, 1, 0, 1$ respectively.

36. In a certain library there are n shelves, each holding at least one book. k new shelves are acquired and the books are rearranged on the $n + k$ shelves, again with at least one book in each shelf. A book is said to be privileged if it is on a shelf with fewer books in the new arrangement than it was in the original arrangement. Prove that there are at least $k + 1$ privileged books in the rearranged library.

37. From a row of $n \geq 12$ consecutive positive integers, two players, first A and then B, take turns crossing out the integer of their choice until there are just two numbers left, a and b. A wins if a and b are relatively prime, and B otherwise.

(i) Show that if n is odd A has a winning strategy.

(ii) Show that if n is even then B has a winning strategy.

38. We say that a finite set $\mathcal{S}$ of points in the plane is *balanced* if, for any two different points A and B in $\mathcal{S}$, there is a point C in $\mathcal{S}$ such that $AC = BC$. We say that $\mathcal{S}$ is *center-free* if for any three different points A, B and C in S, there is no point P in $\mathcal{S}$ such that $PA = PB = PC$.

(a) Show that for all integers $n \geq 3$ there exists a balanced set containing n points.

(b) Determine all integers $n \geq 3$ for which there exists a balanced center-free set containing n points.

39 (Helly's Theorem.)**.** Let $n \geq 4$ convex figures be given in the plane, and suppose each three of them have a common point. Prove that all n figures have a common point.

40. Let $n \geq 4$ points be given in the plane such that each three of them can be enclosed in a circle of radius 1. Prove that all n points can be enclosed in a circle of radius 1.

41 (Jung's Theorem.)**.** Let n points be given in the plane such that each pair of them are at a distance of at most 1 from each other. Prove that all these points can be enclosed in a circle of radius $\frac{1}{\sqrt{3}}$.